PROBLEM

1. DOCTOR SHIFT SCHEDULING USING BACKTRACKING SEARCH

Problem Statement:

A hospital must assign three doctors (D1, D2, and D3) to three shifts (Morning, Afternoon, and Night). Each doctor should be assigned to exactly one shift, and each shift should have only one doctor. The assignment must satisfy the following constraints: D1 cannot work the Night shift, D2 must be assigned to an earlier shift than D3 (Morning < Afternoon < Night), and D3 cannot work the Morning shift. The objective is to use the Backtracking Search algorithm to find a valid shift assignment while checking all constraints at every step.

2.ROBOT PATH FINDING USING SEARCH ALGORITHM

Problem Statement:

A robot operates in a 5 × 5 grid and must move from the Start (S) position to the Goal (G) while avoiding obstacles. The robot can move only up, down, left, and right, with each move having a cost of 1. Obstacles cannot be crossed, and the Manhattan Distance heuristic is used to guide the search. The objective is to determine the optimal path with the minimum cost by showing step-by-step node expansion and priority queue updates.

3. AUTONOMOUS RESCUE ROBOT

Problem Statement:

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The environment is partially observable, meaning the robot can only observe nearby cells. Some paths are blocked by obstacles, and entering risky areas increases the movement cost. The robot must continuously update its knowledge of the environment while searching for the safest and least-cost path to the survivor. The objective is to design an AI-based search strategy that enables the robot to make rational decisions and successfully reach the goal despite limited information and changing conditions.